WORKED SOLUTION

STEP III 2010, Question 3

Pure • Roots of Unity

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 6.1, *Polynomials Whose Roots Are Roots of Unity*: https://beyondstep.keithmcnulty.org/ch06/cyclotomic.html

The key idea

Every root of unity a has a single *order*: the smallest positive m with $a^m = 1$. Being a primitive nth root of unity just means having order exactly n. So we find $C_n(x)$ by listing the nth roots of unity $e^{2\pi ik/n}$, throwing away those that already return to 1 at an earlier power, and multiplying out $(x - a)$ over what is left. The fact that each number has only one order is also what settles the final part.

Solution

Throughout, the nth roots of unity are ω^k for $k = 0, 1, \ldots, n-1$, where $\omega = e^{2\pi i/n}$, and $(\omega^k)^m = 1$ exactly when n divides km. Since C_n is monic with no repeated roots, and its roots are exactly the primitive nth roots of unity,

$$C_n(x) = \prod_{a \text{ primitive}} (x - a).$$

The 4th roots

The 4th roots of unity are $1, i, -1, -i$. Now $1^1 = 1$ and $(-1)^2 = 1$, so these two are not primitive. But $i^1 = i$, $i^2 = -1$, $i^3 = -i$, none equal to 1, and similarly for $-i$ (its powers are the conjugates). So the primitive 4th roots of unity are i and $-i$, and

$$C_4(x) = (x - i)(x + i) = x^2 + 1,$$

as required.

Part (i)

$n = 1$. The only 1st root of unity is 1, and there is no m with $0 < m < 1$, so 1 is primitive: $C_1(x) = x - 1$.

$n = 2$. The square roots of unity are ± 1; 1 is excluded since $1^1 = 1$, while $(-1)^1 \neq 1$. So $C_2(x) = x + 1$.

$n = 3$ **and** $n = 5$. When n is prime, a root ω^k with $1 \leq k \leq n-1$ satisfies $(\omega^k)^m = 1$ only when $n \mid km$, and since n is prime and does not divide k, this forces $n \mid m$. So every nth root of unity other than 1 is primitive. These are the roots of

$$\frac{x^n - 1}{x - 1} = x^{n-1} + x^{n-2} + \cdots + x + 1,$$

which is monic, with distinct roots. Hence

$$C_3(x) = x^2 + x + 1, \qquad C_5(x) = x^4 + x^3 + x^2 + x + 1.$$

$n = 6$. With $\omega = e^{\pi i/3}$, the power ω^k has order $6/\gcd(k, 6)$. So $\omega^0 = 1$ has order 1, $\omega^3 = -1$ has order 2, ω^2, ω^4 have order 3, and only ω and $\omega^5 = \overline{\omega}$ have order 6. Their sum is $2\cos\frac{\pi}{3} = 1$ and their product is $|\omega|^2 = 1$, so

$$C_6(x) = x^2 - x + 1.$$

(As a cross-check, every 6th root of unity has order 1, 2, 3 or 6, so $x^6 - 1 = C_1C_2C_3C_6$, and indeed $(x-1)(x+1)(x^2+x+1)(x^2-x+1) = (x^3-1)(x^3+1) = x^6 - 1$.)

Part (ii)

Let a be any root of $x^4 + 1 = 0$. Then $a^4 = -1$, so $a^8 = 1$. The order d of a divides 8 (write $8 = qd + r$ with $0 \leq r < d$; then $a^r = a^8(a^d)^{-q} = 1$, so $r = 0$ by minimality of d). So $d \in \{1, 2, 4, 8\}$. If d were 1, 2 or 4 then $a^4 = 1$, contradicting $a^4 = -1$. Hence every root of $x^4 + 1$ is a primitive 8th root of unity.

Conversely, ω^k with $\omega = e^{\pi i/4}$ has order 8 exactly when k is odd (if k is even, $(\omega^k)^4 = 1$), so there are exactly four primitive 8th roots: $\omega, \omega^3, \omega^5, \omega^7$. The quartic $x^4 + 1$ has four distinct roots (they are $e^{(2j+1)\pi i/4}$, $j = 0, 1, 2, 3$), all primitive 8th roots, so they are precisely these four. Thus $C_8(x) = x^4 + 1$.

No other n works: the roots of C_n all have order n, and the roots of $x^4 + 1$ have order 8. So the answer is $n = 8$.

Part (iii)

The argument used for $n = 3$ and $n = 5$ in part (i) applies to any prime p: every pth root of unity except 1 is primitive. Therefore

$$C_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1.$$

(For $p = 2$ this gives $x + 1$, agreeing with part (i).)

Part (iv)

Suppose, for contradiction, that $C_q(x) \equiv C_r(x)C_s(x)$ for some positive integers q, r, s.

First, C_s has at least one root, since $e^{2\pi i/s}$ is a primitive sth root of unity: $(e^{2\pi i/s})^m = e^{2\pi i m/s} = 1$ only when $s \mid m$. The same goes for C_r. So C_r and C_s both have degree at least 1.

Take any root a of C_r. By the identity, $C_q(a) = C_r(a)C_s(a) = 0$, so a is a primitive qth root of unity as well as a primitive rth root of unity.

> **Ask yourself** Can one number be a primitive rth root of unity and a primitive qth root of unity with $r \neq q$?

No. If, say, $r < q$, then $a^r = 1$ with $0 < r < q$, which contradicts a being a primitive qth root. If $q < r$, swap the roles. So $r = q$, and in the same way (using a root of C_s) $s = q$.

But then the identity reads $C_q = C_q^2$. Comparing degrees, $\deg C_q = 2 \deg C_q$, so $\deg C_q = 0$, contradicting the fact that C_q has a root. (Alternatively, C_q^2 has repeated roots, which C_q does not.) Hence no such q, r, s exist.

Checks

- The degree of C_n counts the k in $1 \leq k \leq n$ with $\gcd(k, n) = 1$: degrees $1, 1, 2, 2, 4, 2$ for $n = 1, \ldots, 6$, and 4 for $n = 8$, all matching.
- Every nth root of unity has some order d dividing n, so $x^n - 1$ is the product of C_d over the divisors d of n. For $n = 4$: $(x-1)(x+1)(x^2+1) = x^4 - 1$; for $n = 8$: $(x^4-1)(x^4+1) = x^8 - 1$.
- Numerically, multiplying out $(x - a)$ over the primitive roots for $n \leq 40$ reproduces every polynomial above, gives $x^4 + 1$ only for $n = 8$, and finds no identity $C_q = C_r C_s$ with $q, r, s \leq 30$.

Common slips

- Including 1 (or -1) among the primitive roots: C_n is not $x^n - 1$.
- In part (ii), checking only that $x^4 + 1$ divides $x^8 - 1$ without ruling out smaller orders: the point is that $a^4 = -1 \neq 1$.
- In part (iv), arguing from degrees alone: $\deg C_q = \deg C_r + \deg C_s$ can happen (e.g. $4 = 2 + 2$), so the roots, not the degrees, give the contradiction.

Beyond STEP

The C_n are the *cyclotomic polynomials*. The identity $x^n - 1 = \prod_{d \mid n} C_d(x)$ shows they always have integer coefficients, and a deeper theorem says each is irreducible over the

rationals; they are central to Galois theory and to Gauss's discovery of which regular polygons can be constructed with ruler and compass. Allenby's *Rings, Fields and Groups* develops this algebra, and Needham's *Visual Complex Analysis* gives a geometric picture of roots of unity.

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